Veille informatique quantique - 2026-10-09

Période analysée : les 7 derniers jours. Articles retenus : 10.

Synthèse

1. Error-Corrected Memory and Logic on a Heavy-Hex Superconducting-Qubit Processor

Source : arXiv Quantum Physics - publié le 2026-10-09

Scores : pertinence 95/100 | nouveauté 50/100 | importance 50/100 | score 70/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.11658v1 Announce Type: new Abstract: Large quantum computations require high-fidelity error-corrected memory and fault-tolerant logical operations. To achieve these with superconducting circuits, quantum processors will need many components operating with low error rates. In practice, fabrication of monolithic superconducting devices produces unavoidable heterogeneity in component performance, which the quantum computer must accommodate. We implement strategies to operate a large dyn

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computer, qubit, superconducting qubit, quantum error correction, quantum processor.

2. Quantum Networking at the Speed of Quantum Computation: Kilohertz Entanglement in a Heterogeneous Quantum System

Source : arXiv Quantum Physics - publié le 2026-10-09

Scores : pertinence 87/100 | nouveauté 50/100 | importance 50/100 | score 67/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.10705v1 Announce Type: new Abstract: Distributed quantum computing requires rapid, high-fidelity entanglement between qubits in separate processor modules. Photonic links between single-atom qubits provide switchable, long-range connections, but photon loss limits their entanglement rate. Here we entangle a trapped ^{138}Ba^{+} ion with a silicon-vacancy (SiV^{-}) center in a diamond nanophotonic cavity. A photon emitted by the ion is converted in wavelength and encoding, refle

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computing, qubit, trapped ion, quantum networking.

3. Quantum non-Markovian response spectra

Source : arXiv Quantum Physics - publié le 2026-10-09

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.10684v1 Announce Type: new Abstract: Quantum systems are rarely isolated from their surroundings. These hidden environments can retain information and return it later, producing correlated errors in quantum computers and complicating the interpretation of experiments.On a quantum processor, for example, one job can therefore leave an environmental trace that affects the next, even after the qubits themselves have been reset. Learning how this hidden influence persists and affects lat

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Mots-clés détectés : quantum computer, qubit, quantum processor.

4. Geometry-optimized hyperbolic codes for modular fault-tolerant quantum architectures

Source : arXiv Quantum Physics - publié le 2026-10-09

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.10948v1 Announce Type: new Abstract: Quantum fault tolerance is a prerequisite to harness the full potential of quantum computers. However, achieving fault tolerance remains a major challenge since it requires optimizing tradeoffs between several intertwined theoretical and experimental parameters, including the encoding rate, code distance, parity-check weights, code planarity, and routing. Surface codes remain strong candidates for fault tolerance due to their simple geometric stru

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computer, qubit, quantum error correction.

5. Photonic Chips for Universal Quantum Computing: Retrospective and Prospective

Source : arXiv Quantum Physics - publié le 2026-10-09

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.11321v1 Announce Type: new Abstract: Integrated photonic quantum chips have emerged as a promising platform for realizing fault-tolerant photonic quantum computers. Compared with bulk-optics systems, integrated photonics offer compact device footprints, enhanced programmability, and exceptional stability, attributes that are essential for executing large-scale and computationally demanding quantum tasks. In this review, we provide a comprehensive overview of recent advances in both d

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computing, quantum computer, photonic quantum.

6. Beyond QAOA: A Review of AI and Quantum Computing for Adaptive Combinatorial Optimization

Source : arXiv Quantum Physics - publié le 2026-10-09

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.11759v1 Announce Type: new Abstract: Near-term quantum approaches to combinatorial optimization are limited by qubit counts, circuit fidelity, sampling cost, and the difficulty of encoding constraints, while machine learning is increasingly used to configure and control quantum optimization workflows. We call such workflows adaptive: decisions conventionally fixed in advance, from formulation and penalties to shot budgets, backends, and whether to invoke a quantum processor at all, a

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computing, qubit, quantum processor.

7. A quasiparticle-protected superconducting qubit

Source : arXiv Quantum Physics - publié le 2026-10-09

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.12227v1 Announce Type: new Abstract: Superconducting quantum circuits underpin large-scale quantum processors and are increasingly interfaced with optical systems, yet they are vulnerable to quasiparticle poisoning from high-energy radiation and optical fields. Gap engineering of Josephson junctions suppresses quasiparticle-induced relaxation, but quasiparticles near the junction can still shift the qubit frequency and cause phase errors. Here, we extend gap engineering beyond the ju

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : qubit, superconducting qubit, quantum processor.

8. Shortcuts for Adiabatic and Variational Algorithms in Molecular Simulation

Source : arXiv Quantum Physics - publié le 2026-10-09

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2407.20957v2 Announce Type: replace Abstract: Quantum algorithms offer a promising route toward computational advantage, but current implementations remain constrained by limited coherence and gate errors, making circuit complexity and noise sensitivity critical considerations. Here, we develop a shortcuts-to-adiabaticity strategy for molecular ground-state preparation and apply it to both digitized adiabatic quantum computing and the variational quantum eigensolver. In the adiabatic fram

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Mots-clés détectés : quantum computing, quantum processor, quantum algorithm.

9. Scalable Fluxonium-Transmon Architecture for Error Corrected Quantum Processors

Source : arXiv Quantum Physics - publié le 2026-10-09

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2508.09267v2 Announce Type: replace Abstract: We propose a hybrid quantum computing architecture composed of alternating fluxonium and transmon qubits, that are coupled via transmon tunable couplers. We show that this system offers excellent scaling properties, characterized by engineered zero ZZ-crosstalk in the idle regime, a substantial reduction of level-crowding challenges through the alternating arrangement of different qubit types within the lattice, and parameter regimes that ci

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Mots-clés détectés : quantum computing, qubit, quantum processor.

10. Unconditional and exponentially large violation of classicality

Source : arXiv Quantum Physics - publié le 2026-10-09

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2511.11008v2 Announce Type: replace Abstract: Testing the predictions of quantum mechanics has been one of the main experimental endeavors for decades. Recent advancements in technology led to a number of demonstrations which test non-classicality via specific computational tasks. Limitations of these experiments include dependence on complexity theory assumptions, susceptibility to hardware noise and inefficient verification, raising questions about their scalability. We propose to test

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Mots-clés détectés : quantum computer, qubit, Quantinuum.

Méthode

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